

Eightfold — Canonical Candidate Profile Transformer

One-Page Technical Design

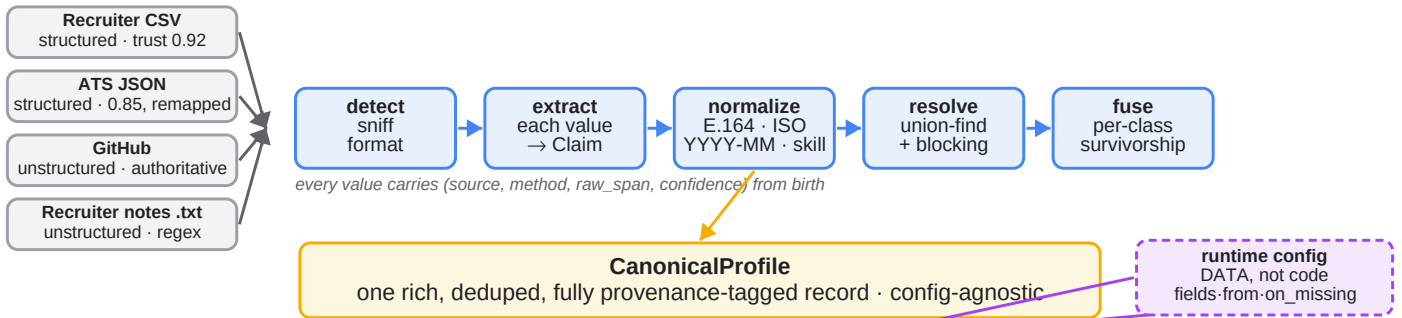
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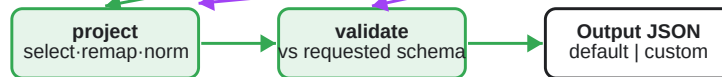
Thesis. Build one **rich, provenance-carrying canonical record** on the **WRITE side**, then a **pure, deterministic projection + validation** layer on the **READ side** — **CQRS for candidate data**. The canonical record is computed *once* and is config-agnostic; the runtime config is *data fed to a generic projector*, never branching logic in the engine. Every value enters as a **Claim{source · method · raw_span · confidence}**, so provenance & confidence are **invariants from birth**. **Governing value:** *wrong-but-confident is worse than honestly-empty* → never invent; abstain to **null**.

Architecture — write once → project per request

WRITE SIDE — built **ONCE**, config-agnostic



READ SIDE — PURE function (CanonicalProfile, Config) → JSON



What it is — and how it differs from the obvious alternatives

- **vs “LLM does the whole transform”:** the steps are *known and fixed* (detect→extract→match→fuse), so a deterministic pipeline wins on the three things hiring needs — reproducible, explainable, hallucination-free. The LLM is **fenced to free prose only**, off by default, cached by input-hash, and emits **low-trust claims that can never overwrite** structured data.
- **vs an agentic loop:** agents earn their keep when the path must be *planned*; record linkage is an auditable fixed sequence — an agent only adds non-determinism, cost & latency for zero correctness gain.
- **vs an ML / embedding entity-matcher:** strong identifiers (email, E.164 phone, normalized URL) are *deterministic* matches — used first; fuzzy name matching is heavily **guarded** (shared block + name-compatible + corroborating signal + no conflicting strong id). Deterministic = explainable + testable.
- **vs best-effort guessing:** an un-normalizable value **abstains to null** (method=normalize_failed, factor 0.0 → it can never win) — a hallucinated employer silently pollutes search; an empty field is honest.
- **vs baking the output shape into merge logic:** CQRS computes truth *once*; a new consumer (search index vs recruiter UI vs export) is a new **config**, not new **code**.

Normalized formats. phones→**E.164** (phonenumbers); dates→**YYYY-MM** (year-only kept, month never fabricated); country→**ISO-3166 a2** (pycountry); skills→**canonical** (swappable ontology). Same registry on write & read sides.

Confidence (explainable). clamp(source_trust × method_factor × normalize_factor + corroboration – conflict, 0, 1); overall = mean of resolved field scores. Every number traces to its inputs.

Merge / survivorship (per field-class). multi-valued→union+dedup; identity/contact→trust-priority winner (csv>ats>linkedin>github>resume>notes); GitHub authoritative for own URL & languages; years_exp→stated, else merged *closed* spans (no “present” → date-independent).

Edge cases. conflicting names→trust picks, both kept in provenance; 3 phone formats→one E.164; malformed/empty source→status, run continues; required + on_missing:error→per-candidate error, no batch crash.

Deliberately descoped (honest scope). sharded / ML-learned matching (single-node blocking + guarded fuzzy *implemented*); full streaming for multi-million-row corpora (–jsonl offered); exhaustive skill ontology; resume PDF/DOCX layout parsing (notes/.txt covers prose); auth / multi-tenancy.

Rationale: maximize the graded core — canonical↔projection separation, provenance/confidence, honest-null — over breadth.